SDG 11 · SUSTAINABLE CITIES & COMMUNITIES

Wait or Walk?

A Smart Commute Decision Assistant for Mumbai's Daily Commuters

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1M1B AI for Sustainability Virtual Internship · IBM SkillsBuild & AICTE

THE REAL PROBLEM

A commute that feels like a daily gamble

A STORY FROM EXPERIENCE

“One day, my connecting bus toward the metro arrived at 6:15 AM. So the next day, I made sure to reach the stop by 6:00 — but the bus never came at 6:15.”

Bus timing near the metro connection isn’t fixed. Some mornings you catch it easily; other mornings you’re stuck guessing — wait a little longer, or start walking?

1

Not a lack of buses

The route exists — the issue is that arrival timing is inconsistent day to day.

2

Past experience misleads

Yesterday’s 6:15 AM bus doesn’t predict today’s. Planning around memory fails.

3

A daily source of stress

Every morning becomes a small gamble — “luck or tension,” as it genuinely feels.

SDG ALIGNMENT & PROBLEM STATEMENT

11

SUSTAINABLE CITIES & COMMUNITIES

Primary SDG

Efficient, predictable urban mobility is core to making cities liveable — last-mile transit friction is a daily quality-of-life issue for millions of commuters.

PROBLEM STATEMENT

Daily commuters — including students — relying on a connecting bus toward the metro need help deciding, in the moment, whether to keep waiting or switch to another option (walking, a different route), because bus arrival times are inconsistent day to day and past experience doesn't reliably predict today.

KEY REFRAME

Not: "When will the bus arrive?" (needs live data we don't have)

Instead: "Should I keep waiting right now, or switch?" — a decision-support problem we can realistically solve.

TARGET USERS

Who this is built for



Students

Commuting daily to college via a connecting bus to the metro — the group this project is grounded in.



Office Commuters

Facing the same rush-hour uncertainty on the way to work, every single morning.



Anyone on the last mile

Anyone bridging the gap between home and the metro without a private vehicle.

The core insight: the problem isn't that buses don't exist — it's that arrival timing is inconsistent, so commuters can't plan with confidence, and every morning becomes a fresh guess.

AI SOLUTION OVERVIEW

A transparent, rule-based decision engine

No public dataset records this specific bus stop's historical lateness — so instead of faking a trained model, we built an honest, explainable decision-support system. It's a legitimate project category listed in the internship guidelines, distinct from a predictive model.

SCORING RULES

Raining today?	**+2**	*grounded in real weather data reasoning*
Weekday rush hour? (8–10am / 6–8pm)	**+1**	*reasoned assumption*
Already waited 10+ min?	**+1**	*reasoned assumption*

Rain is weighted highest — it's the one factor groundable in real, public data (e.g. IMD rainfall records). The others are transparent, reasoned assumptions, clearly labeled as such.

RISK LEVEL → RECOMMENDATION

0–1	**LOW**	*"Wait — you're probably fine."*
2	**MEDIUM**	*"Wait a bit longer, but stay alert."*
3–4	**HIGH**	*"Consider walking or switching routes now."*

PROTOTYPE

How a request flows through the system



WHY THIS DESIGN?

Brain + Delivery split:

The decision logic (“brain”) is fully separate from how it's shown (“delivery”). This keeps the actual AI reasoning simple and testable, while the delivery layer can be a diagram, a chat exchange, or a small interface — all acceptable prototype formats per the guidelines.

Transparent by design:

Every recommendation can be traced back to exactly which factors triggered it — no black-box scoring.

PROTOTYPE / DEMO EVIDENCE

Built with IBM Bob, validated by hand

IBM BOB — GENERATED FUNCTION (simplified view)

```
def commute_decision(rain,
      rush_hour, waited_10min):
    score = 0
    if rain:
        score += 2
    if rush_hour:
        score += 1
    if waited_10min:
        score += 1

    if score <= 1:
        risk = "Low"
        msg = "Wait, you're fine."
    elif score == 2:
        risk = "Medium"
        msg = "Wait, stay alert."
    else:
        risk = "High"
        msg = "Switch routes now."

    return {"score": score,
        "risk_level": risk,
        "recommendation": msg}
```

TEST SCENARIOS (5/5 MATCHED GUT-CHECK)

Scenario	Score	Risk
Mon 8:15am, rain, waited 9 min	3	High
Fri 11am, no rain, just arrived	0	Low
Wed 9am, no rain, waited 10 min	2	Medium
Tue 2pm, rain, just arrived	2	Medium
Sun 8am, rain, waited 10 min	3	High

Each of these 5 morning scenarios was checked by hand before trusting the logic — including a genuinely ambiguous middle case (Wed 9am) — and every result matched real-world intuition.

RESPONSIBLE AI CONSIDERATIONS

Built honestly, with known limits



Assumptions, not measured facts

Thresholds like “10+ min = risky” and “rush hour = busy” are reasoned design choices, not proven from real historical bus-lateness records — because no public dataset tracks this specific stop.



Blind to one-off events

The system only checks rain, rush hour, and wait time. A festival, accident, or road closure could cause real chaos it has no way of detecting — it may still say “Low risk” during genuine disruption.

Future scope: real-time event data, live traffic feeds, and personalization for individual needs (mobility, urgency) — out of scope for this internship’s timeframe, but a clear next step.

EXPECTED IMPACT

What changes if this is used

Commuters and students would feel more confident and less stressed about daily travel, since they'd have a reliable signal on whether to wait or switch — instead of guessing anew every single morning.



Less daily stress

No more guessing — a clear signal replaces daily uncertainty.



Less wasted time

Fewer minutes spent standing uncertainly at the bus stop.



Fewer late arrivals

Better-informed wait/switch decisions, especially on bad-weather days.

Thank You

A small, honest, fully-tested step toward more predictable, less stressful commutes.

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